DG INTERNS HUB — TECHNICAL NOTES

TOPIC 07: EVENT CORRELATION IN SIEM

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1. Concept & Importance

Event correlation connects disparate log entries from multiple network security devices (firewalls, active directory, endpoints, web servers) into a single logical incident story, helping SOC analysts spot complex multi-stage Cyber Kill Chain activities.

2. Correlation Methodologies

- **Rule-Based Correlation:** Evaluates Boolean rules across multiple sources (e.g., Firewall Alert AND Endpoint Malware Alert).
- **Temporal Correlation:** Links events that take place sequentially within a specified time window (e.g., within 5 minutes).
- **Statistical Anomaly Correlation:** Identifies deviations from normal user or host behavioral baselines.

3. Practical SOC Scenario: Correlating Multi-Stage Compromise

> **Scenario: Advanced Spear-Phishing to Data Exfiltration**
>
> **Step 1: Email Gateway:** Inbound phishing email with malicious macro attachment delivered to user `j.doe`.
>
> **Step 2: EDR Telemetry:** `WINWORD.EXE` spawns `powershell.exe` executing encoded script (Event ID 4688).
>
> **Step 3: Firewall Log:** Host `10.0.4.15` initiates 500MB outbound upload to unrated external IP on port 443.
>
> **Correlation Engine Logic:** `If (Phishing Delivery) AND (Process Spawning) AND (Large Outbound Transfer) on SAME Host within 15 Minutes == High Severity Security Incident Alert.`

4. Impact on Alert Fatigue

Without correlation, a SOC team receives hundreds of isolated raw logs. Event correlation consolidates thousands of noisy indicators into a single actionable ticket with full contextual context.